



# A lightweight attention-based framework with multi-scale deep supervision and boundary-aware learning for gastrointestinal tract image segmentation

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## Abstract

Gastrointestinal (GI) tract image segmentation is an important stage in automated diagnostic systems. Segmentation can be assistive and provide information about which area to be segregated. However, the challenges, such as low contrast, irregular shapes, and fuzzy boundaries, make this process challenging. This article presents a lightweight attention-based framework for polyp segmentation. The proposed framework utilizes the EfficientNet-b0 in the encoding phase. A custom decoder is developed that integrates the channel-spatial attention and dynamic gating modules. The attention module enhanced the feature representation, and the dynamic gating module adaptively selects and fuses encoder–decoder features. The auxiliary deep-supervision heads are integrated at the intermediate decoder phases to refine features and stabilize learning. The decoder output is provided to two prediction heads for boundary and mask prediction. A hybrid loss is designed that combines Dice, Lovász-hinge, and Binary Cross Entropy (BCE) loss to optimize region overlap, contour precision, and handle imbalanced classes. For interpretability, Gradient-weighted Class Activation Mapping (Grad-CAM) is incorporated into the proposed framework. The proposed framework provides 94.92% Dice and 90.85% Intersection over Union (IoU) on the Kvasir-Seg, 94.00% Dice and 89.74% IoU on the Hyper-Kvasir, and 93.99% Dice and 88.84% IoU on the CVC-ColonDB dataset. Results on the proposed framework demonstrate significant performance in segmenting the polyps in GI tract images.

**Keywords** Polyp segmentation · Gastrointestinal tract · Deep learning · Deep supervision · Boundary learning

## 1 Introduction

The gastrointestinal (GI) tract is a vital part of the human digestive system. Major parts of the GI tract are the stomach, small and large intestine, and the colon, where mucosal findings differ from extremely mild to fatal diseases. The membrane of mucous has abnormal tissues called polyps [1]. Gastrointestinal polyps are abnormal growths that protrude into the mucosal layer of the GI tract and show a large range of histological and morphological variability [2]. Gastric ulcers occur due to excessive acidity in the stomach.

According to statistics, at any point, 10–20% of the global population is supposed to have encountered an ulcer [3]. The functions of the GI tract are not achieved in the case of GI disorders. GI tract disorders include irritable bowel syndrome, constipation, colon polyps, colitis, cancer, and others. Further, other GI tract lesions, including ulcers and erosions, can also change into cancers, as most GI tract cancers start from the most common lesions provided by endoscopy, called precancerous polyps. Thus, the early recognition as well as removal of GI tract polyps are considered critical for the prevention of GI tract cancers. Many of these can be prevented or minimized by committing to good bowel habits, a healthy lifestyle, and cancer screening tests [4, 5].

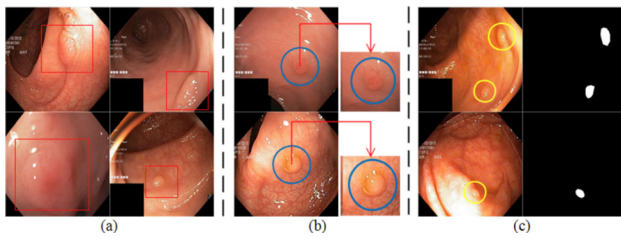
In the US, according to cancer statistics of 2024, the new estimated cases in females and males are 10,730 and 16,160, while the estimated death cases are 4390 and 6490, respectively [6]. Different factors, such as age, health, cancer stage, and cancer type, affect the treatment of GI tract cancer. The common treatment for GI tract cancer includes chemotherapy, radiation therapy, and surgery. The radiation therapy is

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**Fig. 1** Motivation for the proposed framework **a** polyp similarity with healthy background **b** blurred boundaries **c** small-sized polyps

given daily for 15 min for some weeks. With advancements in technology like Magnetic Resonance Imaging (MRI), devices can see the intestine's and tumor's potential daily changes. It is essential to split the organ in the GI tract into segments due to large amounts of X-ray radiation. This technology has the ability to improve the pace of therapy by assisting in the procedure of segmenting the organ of the GI tract [7]. The manual diagnosis is very time-taking and labor-intensive and can be subjective, which introduces intra- and inter-observer errors [8].

Traditional approaches have limited ability to provide the desired level of reliability and precision in GI tract imaging applications [9]. The process of automatic recognition aids the radiologist in achieving effective and prompt cancer recognition, classification, and segmentation [10]. Developments in Artificial Intelligence (AI) and computer vision algorithms have been influenced by various fields of industry and science. These algorithms also simplify various complicated problems in various domains worldwide [11]. In recent years, deep learning frameworks have gained a lot of interest in the domain of segmentation. For the facilitation of CAD, the segmentation is considered a tough and important job [12]. Some recent studies designed methods for segmentation of complex medical images that utilized multi-scale [13] and boundary-aware learning [14]. The GI tract images present challenges such as similarity with healthy background, blurred boundaries, and small-sized polyps. These challenges (see Fig. 1) are not fully addressed by conventional segmentation pipelines. The images used in Fig. 1 are selected from the publicly available datasets that are used for experiments and cited in results section.

In this article, a lightweight attention-based framework is designed for polyp segmentation in GI tract images. An enhanced decoder is introduced, which integrates the Convolutional Block Attention Module (CBAM) and dynamic gating modules. The proposed decoder adaptively combines multi-scale features. Further, a dual-output prediction strategy is employed for accurate polyp boundary estimation. In the intermediate decoder phases, the framework integrates the deep supervision heads for multi-scale feature learning. The effectiveness of these modules relies on the integration

and adaptation to a specific clinical context. Major contribution points are listed below:

- A lightweight architecture is designed for segmentation that consists of an EfficientNet-b0 encoder followed by a custom decoder. The decoder contains CBAM, having channel and spatial attention, and a dynamic gating module to merge encoder features with decoder features.
- The multi-scale auxiliary heads are used in the intermediate decoder phases for deep supervision that performs rich feature learning.
- A dual-prediction strategy is employed that transforms the decoder output in two heads. The output of the first head is for accurate boundary prediction, while the second is for mask prediction.
- Hyperparameters are selected after extensive experimentation such as 16 batch size, 50 epochs, and 0.0002 learning rate. The proposed model is trained from scratch using these hyperparameters.
- A hybrid loss function is designed, which combines Lovász-hinge, Dice, and Binary Cross-Entropy losses. It balances contour precision, pixel optimization, and overlap accuracy, which enhances the segmentation performance. Furthermore, Grad-CAM is applied for the generation of visual explanations that highlight discriminative regions.

## 2 Related work

Researchers have conducted studies that explored disease detection [15, 16] and segmentation of medical images [17]. These studies proposed segmentation frameworks based on Convolutional Neural Networks (CNNs) [18, 19], transformers [20, 21], and hybrid models [22, 23]. In the following, existing polyp segmentation methods are discussed.

### 2.1 Convolutional neural network based methods

Building upon CNNs, research work has been done to improve polyp segmentation [24, 25]. The combination of ResNet50 with DeepLabV3+ was assessed for GI tract polyp segmentation [26]. The proposed combination performed well as compared to pre-trained CNN models. An attention-based stacked dilated convolutional learning residual U-Net model [27] comprised four main phases. First phase includes a multi-level learning process, in the second phase stacked dilated convolutional module is designed, the third phase is an improved convolutional learning module with atrous, in the fourth phase, spatial and channel attention mechanisms having convolutional layers with residual connections. An innovative multi-step framework [28]

was designed for GI tract segmentation in which the first step is preprocessing using intensity normalization and edge enhancement approaches. The segmentation part is initialized by a region-growing algorithm having the advantages of color similarity and spatial coherence. This improves the segmentation while the boundary is precisely extracted through a level set approach. A frequency attention-embedded framework [29] leveraged frequency-based attention mechanisms. The proposed method handles image data into low and high-frequency elements. An intra-component and cross-component attention scheme was integrated that enabled a precise explanation of polyp borders and internal structures. The proposed approach enhanced the polyp segmentation process.

## 2.2 Transformer based methods

Transformer-based approaches are efficient in capturing global context information [30]. By addressing the limitations related to long-range data dependencies, transformer-based models produced improved results for polyp segmentation [31, 32]. ColonGen [33] proposed two transformer models to capture the information from different perspectives. The combined information of both models produced a rich feature representation. ColonGen produced better segmentation outcomes on Kvasir-Seg, CVC-ColonDB, and ETIS-Larib datasets. GSAC-UFormer [34] focused on efficient integration of both global and local details. The fundamental component of the framework is the GSAC-Former block, which fuses CNN-adaptive self-attention with group-wise convolution that enables parallel integration of global and local contexts. The guided attention mechanism for contextual features additionally improves the feature selection process. The technique demonstrated the most appropriate contextual details and refined both channel and spatial relationships in the extracted features. Moreover, the multi-depth partitioned depthwise convolution transformer is utilized as a bottleneck component as it enhances the network learning and optimized feature mapping. A TransUNet [35] was proposed by merging a transformer model with a CNN. The proposed network TransUNet provides remarkable outcomes in segmentation within samples by merging the spatial interpretation of CNNs with the self-attention scheme of transformers. A decoder with multi-head feature receptors [36] was proposed that organizes the encoder features in three groups. Each feature group passed through two three-head feature receptors and one four-head feature receptor. After that, features were fed to the transformer for capturing long-range dependencies. The output of feature heads was considered as auxiliary features, and their prediction and losses were aggregated. The framework outperformed several state-of-the-art techniques on five polyp segmentation datasets.

Across the reviewed approaches, researchers faced several common challenges. Varying GI tract structures, conditions of variable illumination, low contrast, and irregular polyp morphology affect models in accurate polyp segmentation. Blurred and occluded polyp boundaries also degrade the results. To tackle multi-scale feature learning, methods, such as transformers, need massive computational resources. This work introduces an attention-based lightweight framework for accurate polyp segmentation. The proposed framework effectively utilized the attention mechanism, dynamic gating, and deep supervision to address issues such as degraded segmentation results and high computational cost.

## 3 Proposed methodology

The proposed efficient encoder-decoder framework for polyp segmentation is illustrated in Fig. 2. The fine-tuned EfficientNet-b0 [37] is utilized as an encoder. In three blocks of the decoder, the CBAM [38] and a dynamic gating module are integrated. The proposed framework implements multi-scale deep supervision using two heads. Further, the two prediction heads for accurate mask and boundary prediction are employed on the decoder output. The GI tract images of  $256 \times 256 \times 3$  size are inputted into the encoder. The images passed through the encoder-decoder architecture, which efficiently segments the polyp regions. Furthermore, Grad-CAM implemented to visualize the most discriminant features. The Grad-CAM heatmaps are computed from the last convolutional layer.

### 3.1 Encoder architecture

The backbone encoder architecture is based on the EfficientNet-b0. In this encoder architecture, a depth separable convolution of  $3 \times 3$  is employed. On different stages, a stride of 2 is applied, which gradually lowers the spatial size of images, and the number of channels is increased. The model is fine-tuned to adjust the spatial resolution with the decoder. For the given input image  $I(256 \times 256 \times 3)$ , initial feature extraction is performed, and the spatial dimension is reduced to  $128 \times 128$  and then  $64 \times 64$ . After that three feature maps of dimensions  $f1(64 \times 64 \times 64)$ ,  $f2(32 \times 32 \times 128)$ , and  $f3(16 \times 16 \times 256)$ , are produced. The output of these three feature maps is inputted into the decoder blocks. The fourth feature map  $f4(8 \times 8 \times 320)$  extracts fine details at the bottleneck stage. In the bottleneck, convolution is applied using two layers, followed by a Batch Normalization (BN)

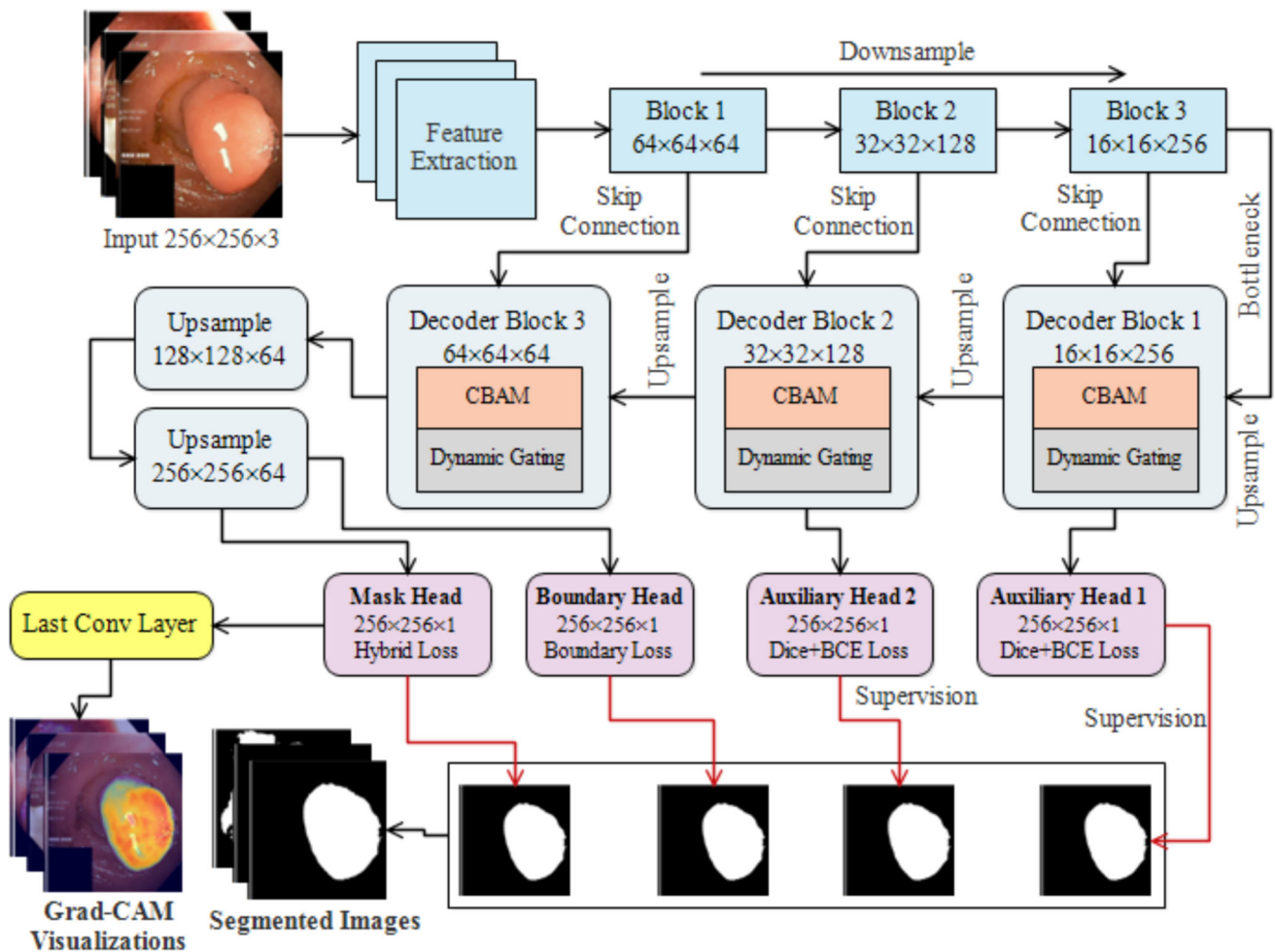


Fig. 2 Proposed framework for polyp segmentation in GI tract images

layer and Rectified Linear Unit (ReLU) activation. This convolutional process in the bottleneck is formulated in Eq. 1.

$$\begin{cases} conv_{b1} = ReLU(BN(conv_{3 \times 3}(f_4)_{320})) \\ conv_{b2} = ReLU(BN(conv_{3 \times 3}(conv_{b1})_{256})) \end{cases} \quad (1)$$

The first convolutional layer employs 320 channels, and the second is applied with 256 channels. While the spatial resolution of  $8 \times 8$  is maintained. The output feature map of  $8 \times 8 \times 256$  is provided to the decoder.

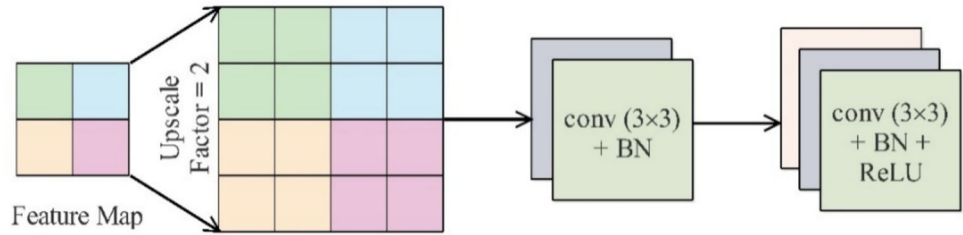
### 3.2 Decoder architecture

The decoder upscales the feature maps to the original spatial resolution. The encoder contextual information is integrated with the decoder blocks of the same resolution. Each decoder block comprises an up-sampling layer with a scaling factor of 2, illustrated in Fig. 3. Two convolutional layers of  $3 \times 3$  are employed with a stride of 1 and padding of 1. In the first convolutional layer, features are normalized with a BN layer.

Second convolution is employed, followed by the BN and ReLU activation layer. After convolutions, two refinement modules, CBAM and dynamic gating, are integrated. At the end, in each decoder block, a dropout layer with a 0.05 ratio is applied.

The bottleneck output  $8 \times 8 \times 256$  is up-sampled  $16 \times 16 \times 256$  and fused with the encoder output  $f_3(16 \times 16 \times 256)$  with a skip connection. In the second decoder block, the feature map  $16 \times 16 \times 256$  is up-sampled to  $32 \times 32 \times 128$  dimensions, and  $f_2(32 \times 32 \times 128)$  encoder output is fused with a skip connection. The third decoder block up-sampled the  $32 \times 32 \times 128$  feature map to the size  $64 \times 64 \times 64$ . The encoder feature map  $f_1(64 \times 64 \times 64)$  is fused via skip connection with the third decoder block. After decoding at three stages, two bilinear up-samplings were performed. The feature map size expanded to  $128 \times 128 \times 64$ , and then  $256 \times 256 \times 64$ . The feature map size matched the input image size for pixel-level prediction.

**Fig. 3** Upscaling and convolutional operation in the decoder block



### 3.2.1 CBAM block

The CBAM block sequentially computes the channel and spatial attention maps. The convolution output of the decoder block is delivered to the CBAM block. First, the channel attention is implemented to compute the channel-wise attention maps. Then, the spatial attention mechanism is employed on the channel attention output that provides the final CBAM output.

For the given input spatial map  $F$  of resolution  $H \times W \times C$ , the channel attention mechanism employed GlobalAveragePooling and GlobalMaxPooling for each channel. After pooling, the size is reduced to  $1 \times 1 \times C$  spatial resolution. This process is formulated in Eq. 2.

$$\begin{cases} P_{avg} = gap(F) \\ P_{max} = gmp(F) \end{cases} \quad (2)$$

After pooling, channel reduction is employed with a reduction ratio of 16, and a ReLU activation operation is performed on the reduced channels. Equation 3 represents the channel reduction process.

$$\begin{cases} C_{avg} = ReLU(P_{avg}(C/16)) \\ C_{max} = ReLU(P_{max}(C/16)) \end{cases} \quad (3)$$

The reduced channels are then reshaped back to the original channels. Then reshaped average and max pooled channels are added, and sigmoid activation is employed as presented in Eq. 4.

$$\sigma_c = Sigmoid(Reshape(C_{avg}) + Reshape(C_{max})) \quad (4)$$

To compute the channel attention output, element-wise multiplication is performed with the input descriptor. Equation 5 describes the channel attention output, which produces the  $H \times W \times C$  size feature map.

$$A_C = F \odot \sigma_c \quad (5)$$

The channel attention output  $A_C$  is sequentially passed to the spatial attention module. This module further refines the channel attention output and extracts fine spatial details.

From the feature map  $A_C$ , channel-wise average and max maps are computed, and a feature map size  $H \times W \times 1$  is obtained. Both average and max maps are concatenated and passed to the convolutional layer. A single convolution filter of  $7 \times 7$  is applied, and the output is given to the sigmoid activation. Finally, to calculate the spatial attention map, the sigmoid output is multiplied by the input feature map  $F$ . Mathematically, the spatial attention process is formulated in Eq. 6.

$$\begin{cases} M_{avg} = Average(A_C) \\ M_{max} = Max(A_C) \\ \{M_{concat} = Concat[M_{avg}, M_{max}]\} \end{cases} \quad (6)$$

$M_{avg}$  and  $M_{max}$  are the average and max maps, respectively.  $M_{concat}$  is the output after concatenating the average and max maps.

$$\sigma_s = Sigmoid(conv_{7 \times 7}(M_{concat})) \quad (7)$$

$$A_S = F \odot \sigma_s \quad (8)$$

where  $\sigma_s$  is the convolution output after applying sigmoid, which is mathematically shown in Eq. 7. In Eq. 8,  $A_S$  is the outcome of the CBAM block, calculated by the multiplication of the input spatial map  $F$  and the sigmoid output  $\sigma_s$ . The combined channel and spatial attention process is demonstrated in Fig. 4.

### 3.2.2 Dynamic gating module

For the regulation of the flow of details between the decoder feature map and encoder skip connection, the proposed architecture integrates a dynamic gating module. A learnable gating scheme, which is parameterized by a sigmoid-weighted tensor, is employed. This tensor balances the contribution from both decoder features and encoder features, which enables the network to block irrelevant details and demonstrate task-related structures. The gate is activated with a balance prior to 0.5, which ensures unbiased learning at the initial stage of training. Further, it progressively learns the fused optimal weights through backpropagation. Formally, let  $E \in S^{B \times C \times H \times W}$  present the skip connection from the



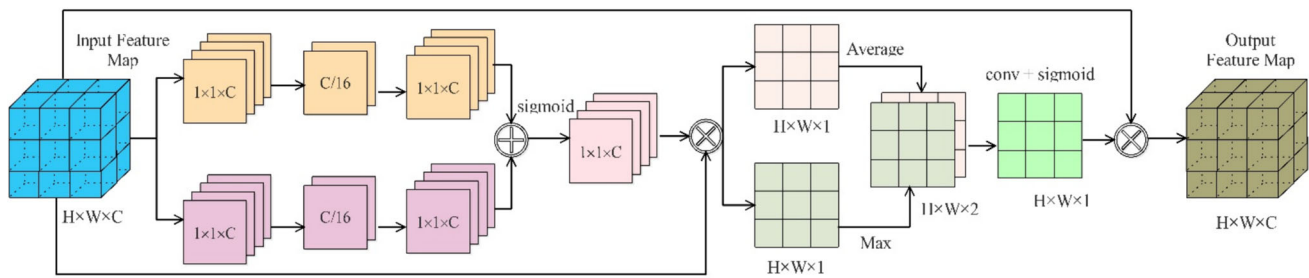


Fig. 4 CBAM block for channel and spatial information learning

encoder and similarly,  $D \in S^{B \times C \times H \times W}$  for decoder features at the same level of resolution. The learnable parameter  $\beta \in S^{1 \times C \times 1 \times 1}$  transformed using the sigmoid activation function to get the gating mechanism as shown in Eq. 9.

$$O = \xi(\beta) \quad (9)$$

where  $\xi(\cdot)$  shows the element-wise sigmoid function. The fused computed output is shown in Eq. 10.

$$Fused = O \odot S + (1 - O) \odot D \quad (10)$$

where  $\odot$  shows element-wise multiplication. The design component performs a convex emergence of decoder features and skip connections, through a getting tensor  $O$ , which determines the contribution in each pathway for every channel.

### 3.3 Deep supervision

The proposed framework integrates the auxiliary deep-supervision heads at the intermediate phase of the decoder. In conventional approaches, the output of the decoder contributes to the loss, which may generate weak gradients and slow convergence. This issue is addressed using the auxiliary branches at the decoder level. Keeping in view the computational cost, the auxiliary head is implemented as a lightweight convolution of size  $1 \times 1$ . Auxiliary head maps the intermediate attributes to the preliminary segmentation logits, then upscales to the original input image resolution. This scheme aims that the intermediate layer to generate meaningful semantic predictions. The feature map generated by each head is mathematically shown in Eq. 11.

$$A_i = \text{Upsample}(\text{Conv}_{1 \times 1}(B_i), \text{Size} = (H, W)) \quad (11)$$

where  $B_i$  denotes an intermediate features map and  $H, W$  shows the spatial size of image. Finally, a weighted sum of the auxiliary losses is then introduced as  $\mathcal{L}_{aux,i}$  and  $\mathcal{L}_{main}$

as shown in Eq. 12.

$$\mathcal{L}_{total} = \sum_{i=1} \zeta_i, \mathcal{L}_{aux,i} + \mathcal{L}_{main} \quad (12)$$

where  $\zeta_i$  are the small weight factors. These factors contribute in auxiliary heads learning without dominating the optimization process. The integration of deep supervision improves training efficiency. The predictive capability of the decoder at multiple depths helps the framework to learn significant multi-scale representations.

### 3.4 Dual-prediction strategy

The decoder output is provided to two prediction heads. The output of dimension  $256 \times 256 \times 64$  is further transformed into two outputs of size  $256 \times 256 \times 1$ . Both outputs were utilized for polyp prediction using the dual-prediction approach. One prediction head is used for mask prediction, and the other is used for accurate boundary estimation.

## 4 Experimental results and discussion

All the experiments are conducted on Kaggle using a GPU P100. The metrics used to evaluate the framework include Dice coefficient, Intersection over Union (IoU), precision (Pr), recall (Rc), and accuracy (Acc). Experiments are conducted on three datasets, including Kvasir-Seg [39], Hyper-Kvasir [40], and CVC-ColonDB [41]. The data is distributed into training and testing sets. The training set utilized 70% of the data, and 30% data is utilized for testing purposes. The training and testing splits are performed at the image level. For the training, data augmentation techniques are utilized, such as image scaling, horizontal flipping, vertical flipping, adding a Gaussian blur, and image sharpening.

### 4.1 Loss function and hyperparameters

A composite loss function is formulated to enhance the pixel-level accuracy and to optimize the polyp boundary segmentation. The designed hybrid loss function is built upon the

integration of the Dice loss [42], BCE loss [43], and Lovász hinge loss [44]. For the given target label  $g$  and output mask  $\hat{g}$ , the Dice loss is defined in Eq. 13.

$$\begin{cases} \text{Dice}(g, \hat{g}) = \frac{2\sum_i g_i \hat{g}_i}{\sum_i g_i^2 + \sum_i \hat{g}_i^2} \\ \ell_{\text{Dice}} = 1 - \text{Dice}(g, \hat{g}) \end{cases} \quad (13)$$

The BCE loss for each pixel  $i$ , is defined in Eq. 14.

$$\ell_{\text{BCE}} = -\frac{1}{N} \sum_i [g_i \log \hat{g}_i + (1 - g_i) \log (1 - \hat{g}_i)] \quad (14)$$

Lovász hinge loss for the binary masks is formulated in Eq. 15.

$$\ell_{\text{Lovasz}} = \frac{1}{|K|} \sum_{k \in K} \overline{\Delta J}(s(k)) \quad (15)$$

where  $\overline{\Delta J}$  are the gradient values of IoU, and  $s(k)$  are the sorted pixel errors. For training and testing, the hybrid loss is defined, which handles the class imbalance, pixel-level accuracy, and optimizes the IoU. The combined loss function contains the following weightages: 0.5 Dice loss, 0.15 Lovász hinge loss, and 0.35 BCE loss. The mentioned weights are determined by an experimental tuning process, as extensive experiments are conducted, which consist of different combinations, and at last, final weights are selected based on consistent and stable performance. The ablation results of hybrid loss weights are provided in Supplementary Material 1. Hybrid loss  $\mathbb{L}$  is mathematically defined in Eq. 16.

$$\mathbb{L} = 0.5 \times \ell_{\text{Dice}} + 0.35 \times \ell_{\text{BCE}} + 0.15 \times \ell_{\text{Lovasz}} \quad (16)$$

The model is trained with 16 batch size, in 50 epochs, having initial learning rate of 0.0002. Cosine Annealing with Warm Restarts is utilized to optimize the learning rate after every epoch, which maintains stability and avoids premature convergence. During the training process, AdamW is used as the network optimizer.

## 4.2 Segmentation results and comparisons

This section describes the quantitative outcomes as well as comparisons with the existing techniques. The proposed method outperformed various existing methods and achieved better polyp segmentation results. Evaluating the proposed method on the Kvasir-Seg, a Dice of 94.92% and an IoU of 90.85% are obtained. Accuracy, precision, and recall achieved on the Kvasir-Seg are 98.49, 93.26, and 97.39%, respectively. Using the Hyper-Kvasir dataset of polyp segmented images, the proposed method attained 94.00% Dice and 89.74% IoU. On the Hyper-Kvasir dataset, accuracy is

98.33%, precision is 92.79%, and recall is 95.82%. Dice and IoU calculated on the CVC-ColonDB dataset are 93.99 and 88.84%, respectively. The calculated accuracy is 99.41%, precision is 91.88%, and recall is 96.45%.

Table 1 presents the comparison with existing techniques utilizing the Kvasir-Seg dataset. The developed model outperformed the results reported in Diffusion Text-Attention Network [24], MultiResUNet [25], Polyp-LVT [30], CAFÉ-Net [31], BUNet [32], ColonGen [33], MSEANet [45], ConvNeXt model [46], MSBP-Net [47], and WBANet [48] in terms of Dice, IoU, accuracy, and recall. The precision achieved using the proposed framework is also better in comparison with other methods, but the reported precision in MultiResUNet [25] is slightly better. As shown, the proposed method achieved a precision of 93.26% and MultiResUNet [25] reported 93.64%.

As shown in Table 2, the Diffusion Model [23] achieved 80.42% Dice and 73.96% IoU on the Hyper-Kvasir dataset. In [26], DeepLabV3+ with ResNet-50 as backbone architecture reported 86.04% Dice, 97.01% accuracy, 91.68% precision, and 86.98% recall. The proposed method outperformed the existing approaches in terms of all selected metrics.

Table 3 shows the comparison of the proposed framework using the CVC-ColonDB dataset. As shown, the existing methods calculate only Dice and IoU, but the proposed method is validated with three additional metrics accuracy, precision, and recall. Dice and IoU reported in CAFÉ-Net [31], BUNet [32], ColonGen [33], MSBP-Net [47], and WBANet [48] are 83.20% and 75.20%, 81.30% and 73.10%, 81.90% and 74.00%, 81.00% and 73.10%, 82.30% and 74.60%, respectively. The dice is 10% improved on the CVC-ColonDB, which shows the model's ability to handle the complex characteristics of this dataset. The proposed model consists of attention mechanism, deep supervision, and dynamic gating modules. These modules mutually enhance boundary sensitivity and robustness, that provides substantially improved segmentation performance.

Table 4 presents a comparison with established segmentation models that are implemented on the Kvasir-Seg dataset with the image resolution size of  $256 \times 256$ . The proposed model compares the metrics, including frames per second (FPS), number of training parameters, and Giga floating point operations (GFLOPs). The proposed model consists of 5.20 million parameters, and 14.62 GFLOPs with 62.01 FPS inference time. The proposed model produced better segmentation results as compared to other models with a decreased computational cost.

The visual segmentation results on the Kvasir-Seg, Hyper-Kvasir, and CVC-ColonDB datasets are displayed in Fig. 5. It can be seen in the figure that the results of the proposed framework are more precise than other established methods. The proposed framework captured accurate boundary information and efficiently segmented the small and large polyps.

**Table 1** Comparison of quantitative results using the Kvasir-Seg dataset

| Methods                                | Year | Dice (%) | IoU (%) | Acc (%) | Pr (%)       | Rc (%) |
|----------------------------------------|------|----------|---------|---------|--------------|--------|
| Diffusion Text-Attention Network [24]  | 2024 | 90.39    | 84.06   | –       | 92.01        | 91.56  |
| MultiResUNet [25]                      | 2024 | 86.63    | 82.77   | 95.93   | <b>93.64</b> | 80.60  |
| Polyp-LVT [30]                         | 2024 | 90.90    | 85.10   | –       | –            | –      |
| CAFÉ-Net [31]                          | 2024 | 92.80    | 87.30   | –       | –            | –      |
| BUNet [32]                             | 2024 | 92.30    | 87.30   | –       | –            | –      |
| ColonGen [33]                          | 2024 | 93.00    | 88.40   | –       | –            | –      |
| MSEANet [45]                           | 2025 | 91.92    | 86.91   | –       | 93.87        | 92.45  |
| ConvNeXt model, Transformer block [46] | 2025 | 88.02    | 81.20   | –       | 91.00        | 89.50  |
| MSBP-Net [47]                          | 2025 | 91.90    | 86.80   | –       | –            | –      |
| WBANet [48]                            | 2025 | 93.30    | 88.90   | –       | –            | –      |
| Proposed                               |      | 94.92    | 90.85   | 98.49   | 93.26        | 97.39  |

Bold values indicate the better results against each performance measure

**Table 2** Comparison of quantitative results using Hyper-Kvasir's polyp images dataset

| Methods                      | Year | Dice (%) | IoU (%) | Acc (%) | Pr (%) | Rc (%) |
|------------------------------|------|----------|---------|---------|--------|--------|
| Diffusion Model [23]         | 2024 | 80.42    | 73.96   | –       | –      | –      |
| DeepLabV3 + , ResNet-50 [26] | 2024 | 86.04    | –       | 97.01   | 91.68  | 86.98  |
| Proposed                     |      | 94.00    | 89.74   | 98.33   | 92.79  | 95.82  |

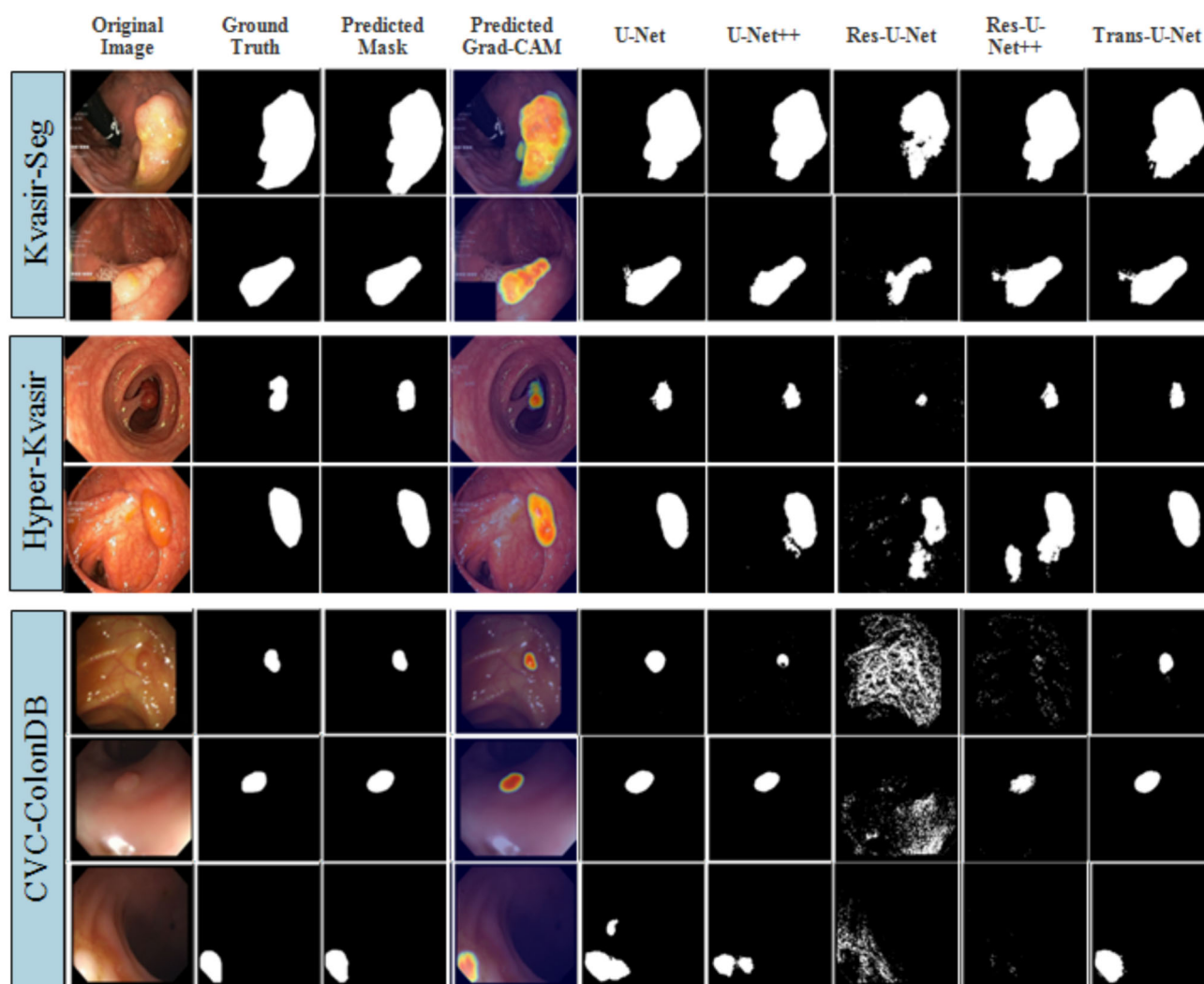
**Table 3** Comparison of quantitative results using the CVC-ColonDB dataset

| Methods       | Year | Dice (%) | IoU (%) | Acc (%) | Pr (%) | Rc (%) |
|---------------|------|----------|---------|---------|--------|--------|
| CAFÉ-Net [31] | 2024 | 83.20    | 75.20   | –       | –      | –      |
| BUNet [32]    | 2024 | 81.30    | 73.10   | –       | –      | –      |
| ColonGen [33] | 2024 | 81.90    | 74.00   | –       | –      | –      |
| MSBP-Net [47] | 2025 | 81.00    | 73.10   | –       | –      | –      |
| WBANet [48]   | 2025 | 82.30    | 74.60   | –       | –      | –      |
| Proposed      |      | 93.99    | 88.84   | 99.41   | 91.88  | 96.45  |

**Table 4** The proposed model's results comparison with existing models using the Kvasir-Seg dataset

| Models            | Parameters (Million) | GFLOPs | Inference time (FPS) | Dice (%) | IoU (%) |
|-------------------|----------------------|--------|----------------------|----------|---------|
| U-Net [49]        | 34.52                | 65.44  | 139.86               | 78.29    | 69.74   |
| U-Net + + [50]    | 36.63                | 1380   | 67.53                | 78.22    | 69.53   |
| Res-UNet [51]     | 13.04                | 71.21  | 119.91               | 62.80    | 51.76   |
| Res-UNet + + [52] | 14.48                | 70.77  | 99.69                | 70.70    | 61.26   |
| ACSNet [53]       | 29.45                | 16.33  | 102.59               | 89.39    | 84.10   |
| TransUNet [54]    | 93.20                | 53.40  | 125.00               | 86.70    | 79.50   |
| Proposed          | 5.20                 | 14.62  | 62.01                | 94.92    | 90.85   |





**Fig. 5** Test images and corresponding segmentation results by the proposed and other established models

Moreover, Grad-CAM visualizations on the proposed framework highlight the potential for clinical practices.

### 4.3 Ablation study

The ablation analysis is carried out on the Kvasir-Seg dataset to validate the effectiveness of the proposed modules. The effectiveness of the proposed core architectural modules, such as CBAM and dynamic gating, is analyzed by selectively removing them from the network.

Table 5 summarizes the results of ablation experiments. Removing the dynamic gating module, the Dice and IoU reduced by 1.70 and 2.10%, respectively, while excluding the CBAM, the Dice and IoU dropped by 2.40 and 3%, respectively. The model obtained even worse outcomes when both modules are removed, as both Dice and IoU dropped by 4.50 and 5.20%, respectively. The significant drop in performance demonstrated that both modules, dynamic gating and CBAM,

efficiently performed in accurate polyp segmentation, as the addition of both modules improves the boundary refinement and feature discrimination.

To deploy the framework in real-time scenarios, computational efficiency is an important factor. This trade-off is considered in the designed architecture by merging a lightweight encoder with attention, deep supervision, and dynamic gating. The framework obtains significant segmentation results with fewer parameters than heavier architectures, that make it well-suited for practical and real-time implementation. The proposed framework has the potential to be integrated into real-time endoscopic systems, which can help in rapid and accurate GI tract polyp segmentation and assist in clinical decision-making. The model is computationally efficient as compared to other models, but a more lightweight or compressed version of this model can be integrated into portable devices. Although better segmentation results are achieved on selected datasets, however, this research work can be

**Table 5** Ablation studies for the proposed architecture

| Dataset    | Architecture |                | Dice (%) | IoU (%) |
|------------|--------------|----------------|----------|---------|
|            | CBAM         | Dynamic gating |          |         |
| Kvasir-Seg | ✓            | ✓              | 94.92    | 90.85   |
|            | ✓            | ×              | 93.22    | 88.95   |
|            | ×            | ✓              | 92.52    | 87.85   |
|            | ×            | ×              | 90.42    | 85.65   |

enhanced in the future to achieve generalizability across other endoscopic systems or patients from different regions. More multi-center publicly available datasets, such as PolypGen, CVC-ClinicDB, and ETIS-LaribPolypDB, can be utilized for evaluation across diverse patient groups. Since data splitting is conducted at the image level, images from the same patient or video sequence may appear in the training and test sets. Furthermore, clinical interpretability can be improved by integrating more explainability modules.

## 5 Conclusion and future work

This work proposed a deep learning approach for the accurate polyp segmentation in GI tract images. The framework employs EfficientNet-b0 as an encoder. A decoder is designed that incorporates CBAM and dynamic gating modules. The encoder backbone provided multi-scale features while the model remained computationally efficient. The decoder comprised a sequential attention mechanism that helped to emphasize the most informative polyp regions and improved the localization results. The dynamic gating module adaptively balanced the decoder and skip connection features. The adaptive behavior preserved the fine-grained spatial details. The intermediate decoder layers employed deep supervision through auxiliary output heads. Deep supervision enhanced multi-scale feature learning and improved the gradient flow. The enhanced feature learning and gradient propagation refined the model training process. For polyp boundary refinement and prediction, the dual-prediction strategy performed well and produced more accurate segmentation results. Furthermore, a hybrid loss balanced the global and local boundary precision and led to the accurate polyp boundary prediction. The model precisely segments the small-sized and irregular polyps. In the future, models will be developed and evaluated on more datasets. This will help to produce a more generalized model that will be applicable to diverse clinical conditions.

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**Data availability** The datasets Kvasir-Seg, Hyper-Kvasir, and CVC-ColonDB utilized in this study are publicly available.

## Declarations

**Conflict of interest** The authors declare no competing interests.

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